

# AI Gmail Email Classifier

Google Apps Script + OpenAI automation for business inbox triage

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| <b>Type</b><br>Portfolio demo | <b>Role</b><br>AI Automation Developer | <b>Stack</b><br>Google Apps Script, Gmail, OpenAI, Google Sheets |
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## Problem

Business inboxes often mix invoices, customer requests, complaints, order updates, marketing messages, and internal notes. Without automation, a team member has to manually scan and sort messages before the right person can act.

## Solution

I built a Google Apps Script workflow that checks unread Gmail messages, sends a short email summary to OpenAI for classification, applies the correct Gmail label, marks the message as processed, and logs errors to Google Sheets.

## Workflow

|                                             |                                                            |                                                              |                                                    |
|---------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------|
| <b>1. Gmail</b><br>Read unread inbox emails | <b>2. OpenAI</b><br>Classify subject, sender, body snippet | <b>3. Gmail Labels</b><br>Apply category and processed label | <b>4. Google Sheets</b><br>Log failures and errors |
|---------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------|

## Key Features

- Time-driven automation that can run every 5 minutes
- Batch processing to stay within Google Apps Script execution limits
- Configurable category list for invoices, orders, complaints, quote requests, marketing, internal messages, and other
- OpenAI JSON classification with low-confidence fallback handling
- Processed marker so messages are not classified twice
- Google Sheets error log with timestamp, email subject, and error message
- API key stored in Script Properties instead of being hardcoded
- Setup documentation for labels, trigger, API key, and deployment

## Implementation Snapshot

The project is intentionally lightweight: no external hosting, no database server, and no Node.js deployment. Everything runs inside Google Apps Script and uses Gmail, Google Sheets, and OpenAI.

```
function classifyUnreadEmails() {
  const unreadThreads = GmailApp.search(
    "in:inbox is:unread -label:" + quoteGmailLabel_(CONFIG.PROCESSED_LABEL),
    0,
    CONFIG.MAX_EMAILS_PER_RUN
  );

  unreadThreads.forEach(function(thread) {
    const message = thread.getMessage().slice(-1)[0];
    const emailInput = buildEmailInput_(message);
    const classification = classifyEmailWithOpenAI_(emailInput);
    const category = resolveCategory_(classification);

    thread.addLabel(getOrCreateLabel_(category.label));
    thread.addLabel(getOrCreateLabel_(CONFIG.PROCESSED_LABEL));
  });
}
```

## Business Value

- Reduces manual inbox triage and repetitive sorting work
- Helps important emails reach the right workflow faster
- Can be adapted for support routing, lead sorting, invoice detection, quote requests, and admin operations
- Keeps the setup simple enough for small business teams to understand and maintain

## Portfolio Note

This is a portfolio demo, not a live client deployment. It uses realistic production patterns such as Script Properties, batch limits, fallback labels, error logs, and setup documentation so it can be adapted quickly for a real Gmail automation project.